

DR6e: Claude Caught the Novel Realisation – Full 2x2 Confirmed

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Deletion-Repair – DR6e **Status:** Overall **G0**. U6e **G0** (R6 consensus 10/10), W6 **NO_G0** (overlap gap +7). **Wall absent under open enumeration with semantic reasoning.** Completes the 2x2 truth table on DR5's sharpened antecedent. **Date:** 2026-07-27

Abstract

DR6d hit the wall using a regex verifier on an extended snippet set with one novel realisation R6. Sharpened DR5 predicted a Claude verifier on the same 11 snippets would NOT hit the wall, because condition (b) (semantic reasoning) escapes even when condition (a) (open enumeration) fails. DR6e ran that test.

Result: three Claude verifiers all scored R6 = 10. R6 was caught semantically at the same strength as R1 and R5 (also 10). Realisation distribution [10, 9, 8, 8, 10, 10], placebo distribution [1, 0, 0, 0, 0]. Overlap gap +7. **U6e G0. W6 NO_G0. Wall absent.**

The full 2x2 truth table on the sharpened DR5 antecedent:

(a) enumeration open at test time?	(b) semantic reasoning available?	wall?
yes	no	YES (DCR1f physics, DR6d code)
yes	yes	NO (DR6e)
no	no	NO (DR6c)
no	yes	NO (DR6)

The wall bites if and only if both antecedents fail. This is the theorem operating exactly as DR5 claims, empirically confirmed on all four corners across two subject-matter domains and two verifier architectures.

1. Setup

Same 11 snippets as DR6d (5 original realisations + 5 placebos + R6 novel). Three sandboxed Claude subagents, blind to each other and to labels. Prompt byte-identical to DR6 except for pointing to the extended snippet file. Consensus median.

2. Results

Per-snippet Claude consensus scores:

snippet	kind	per verifier	consensus
R1_utcnow_direct	realisation	10 / 10 / 10	10
R2_replace_tzinfo_none	realisation	9 / 9 / 9	9
R3_time_fromtimestamp	realisation	8 / 8 / 8	8
R4_iso_parse_no_tz	realisation	9 / 8 / 7	8
R5_combine_utc_convention	realisation	10 / 10 / 10	10
R6_json_field_no_tz	realisation	10 / 10 / 10	*10*
P1_pytz_localize	placebo	1 / 1 / 1	1
P2_now_with_timezone_utc	placebo	0 / 0 / 0	0
P3_zoneinfo_user_tz	placebo	0 / 0 / 0	0

P4_no_datetime_arithmetic	placebo	0 / 0 / 0	0
P5_arrow_aware	placebo	0 / 0 / 0	0

Aggregate:

- All 6 realisations (including R6): median 9.5, stdev 0.98, min 8, max 10.
- All 5 placebos: median 0, max 1.
- Overlap gap = min(realisation) - max(placebo) = 8 - 1 = **+7**.

3. Gate decisions

gate

U6e R6 caught semantically	G0	consensus 10, threshold 6
W1 verifier completeness	G0	
W2 realisation median ≥ 6	G0	median 9.5
W3 realisation stdev ≥ 1.5	NO_G0 (as expected)	0.98 – realisations are uniform-good
W4 placebo median ≤ 3	G0	median 0
W5 placebo trigger	NO_G0	max placebo 1, no projection
W6 realisation-placebo overlap	NO_G0	gap +7

Overall G0 on the DR6e question: U6e G0 (semantic catch) + W6 NO_G0 (no overlap) = wall absent under open enumeration with semantic reasoning. Licensed reading:

wall_absent_under_open_enumeration_with_semantic_reasoning: sharpened
DR5 confirmed on the (a)-open, (b)-satisfied corner. Claude caught R6
semantically. Escape via condition (b) is real.

4. What DR6e establishes

R6's surface form was not represented in the DR6c regex label set; the regex verifier scored R6 = 0 (DR6d wall). Claude scored R6 = 10 by every verifier. The mechanism is semantic reasoning about \$D\$: Claude read the code, understood that datetime(year=..., month=..., ...) constructs a naive datetime interpreted by convention as UTC, and matched that against \$D\$'s natural-language description.

All four corners empirically observed. Wall bites iff (a) yes AND (b) no – exactly what the theorem's antecedent specifies.

regex matcher) and DR6d (2026 Python, regex matcher) both hit the wall at the same corner. DR6 (Python, Claude), DR6c (Python, regex enum-closed), DR6e (Python, Claude enum-open) all escaped at the respective corners DR5 predicted.

- **Semantic reasoning does escape the wall under open enumeration.**
- **The 2x2 truth table on DR5's sharpened antecedent is now complete.**
- **Cross-corpus, cross-architecture agreement.** DCR1f (1900 physics,

5. What DR6e does not license

not reach R6 semantically. A different novel realisation (say, binary-encoded UTC timestamp deserialisation) may not be as recognisable to Claude.

remains a structural claim; the empirical corner just confirms it. Class-aware nomination with a sound \$g\$ remains DR5's structural escape.

a different experiment. Withholding $\$D\$$ may or may not preserve semantic reasoning; DR6f would test.

- **All LLM verifiers escape under all conditions.** Weaker LLMs may
- **DR5 is inescapable in the (a)-yes, (b)-no corner.** The theorem
- **DR6f is now decided.** DR6f (Claude WITHOUT $\$D\$$ in the prompt) is

6. Next follow-ups

Predicted: the wall reappears because the verifier cannot access the canonical form without knowing $\$D\$$.

requires an external $\$g\$$, what soundness/completeness properties must $\$g\$$ have to be trusted for $\$D\$$?

now that DCR2a's cardinality-based rules were shown insufficient at the placebo.

- **DR6f** – Claude verifier where $\$D\$$ is not stated in the prompt.
- **DR7** – grouping-function correctness. If class-aware nomination
- **DCR2b** – multi-document coverage aggregation on the DCR corpus,

Appendix: reproduction

```
uv run --no-sync python -m experiments.dr6_code_correctness_corollary.run_dr6e
```

Reads three verifier JSON files (verifier_ext_A/B/C.json), aggregates by median, applies the six W-gates plus U6e, writes results/dr6e_verdict.json.

Preregistration **digest** **(SHA-256** **of** DR6E_PREREGISTRATION.md):
4a7e7720fd629a6eb85883e33e43bb80dd16038e4fb9d2c0a7e5a5645fb96402.